

JP4

JP2

D1

D2

PWR

GND

C4 C5

U1

R2

R1

C3

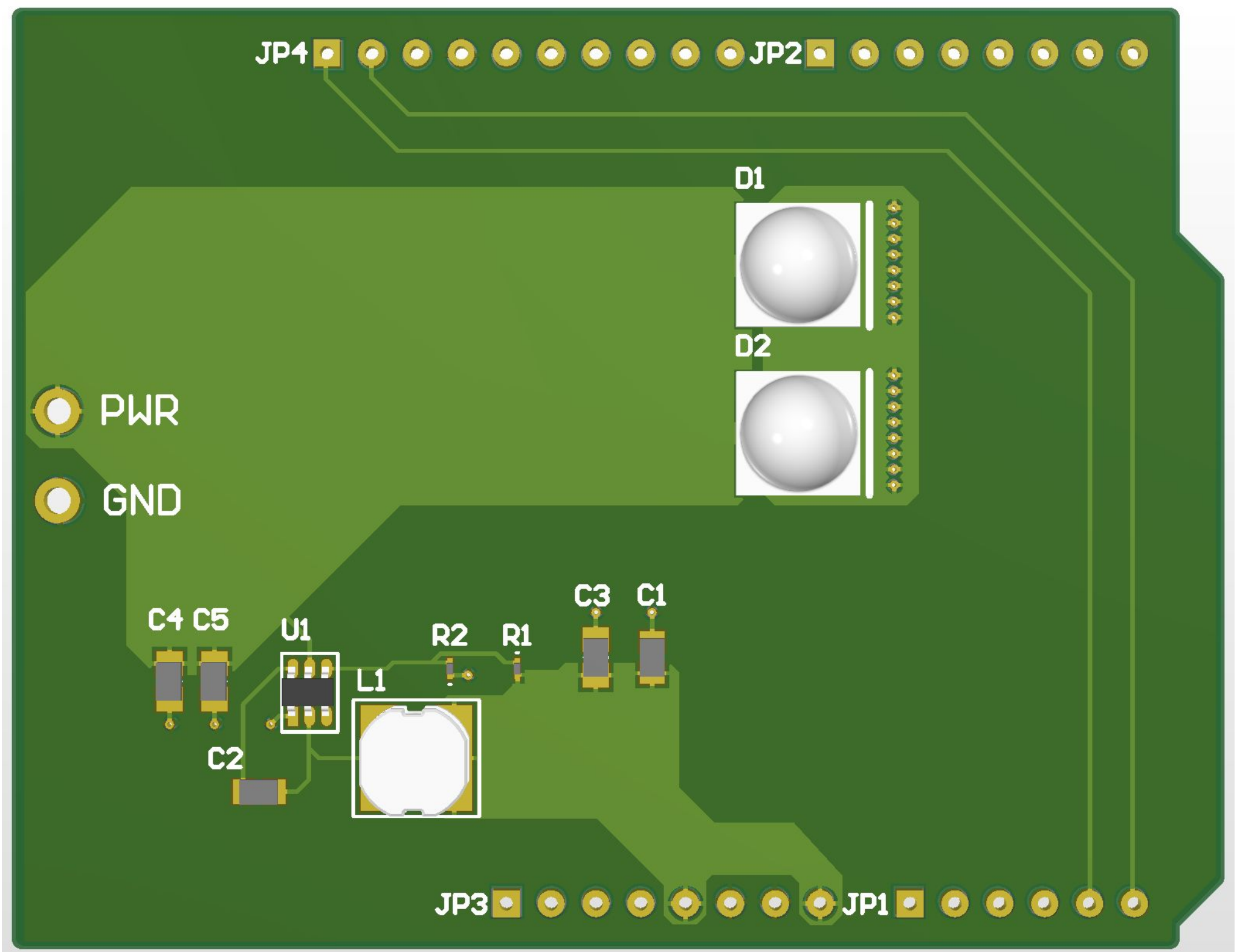
C1

L1

C2

JP3

JP1



JP4

JP2

D1

D2

PWR

GND

C4 C5

U1

R2

R1

C3

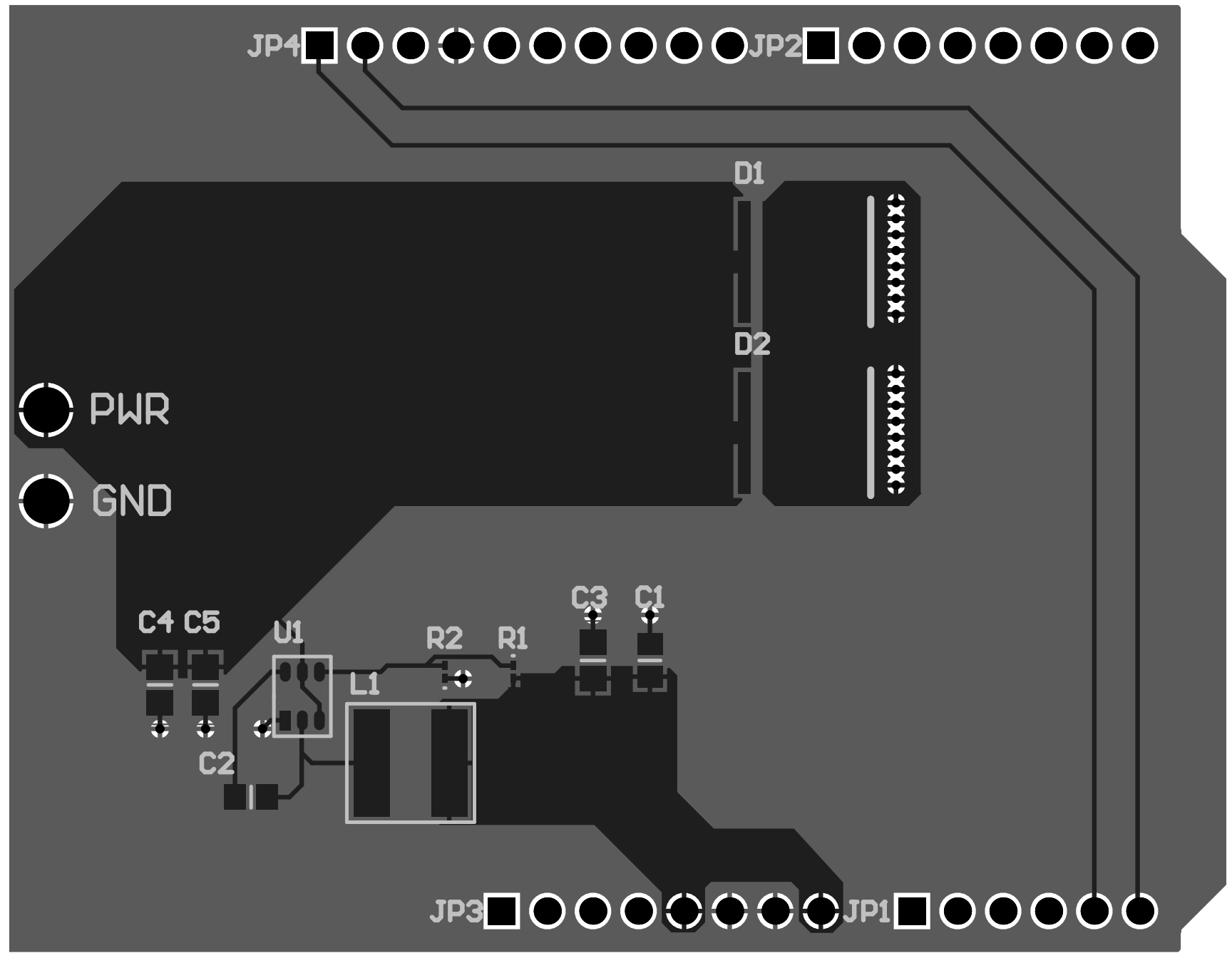
C1

L1

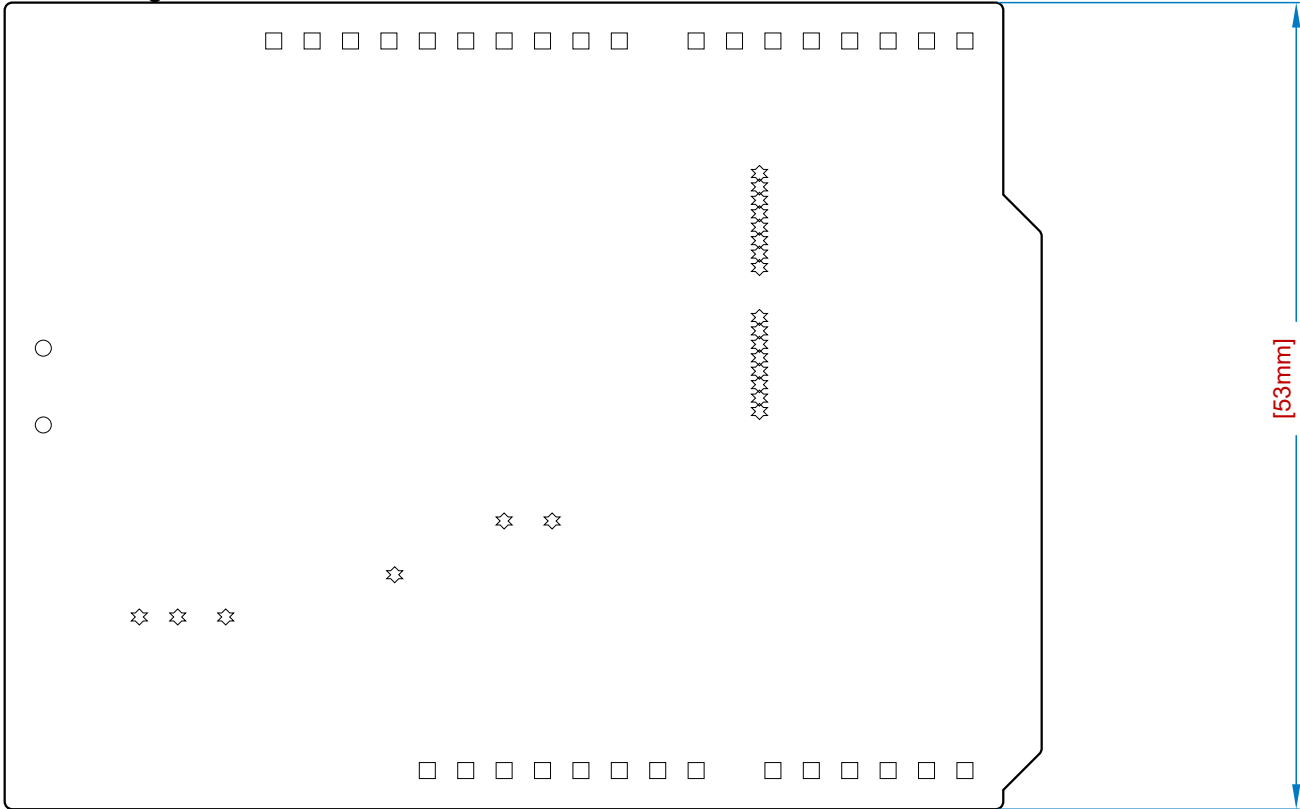
C2

JP3

JP1



Drill Drawing View



Layer Stack Legend

Material	Layer	Thickness	Dielectric Material	Type	Gerber
	Top Overlay			Legend	GTO
Surface Material	Top Solder	0.01mm	Solder Resist	Solder Mask	GTS
Copper	Top Layer	0.04mm		Signal	GTL
		1.52mm	FR-4	Dielectric	
Copper	Bottom Layer	0.04mm		Signal	GBL
Surface Material	Bottom Solder	0.01mm	Solder Resist	Solder Mask	GBS
	Bottom Overlay			Legend	GBO
Total thickness: 1.62mm					

Drill Table

Symbol	Count	Hole Size	Plated	Hole Tolerance
○	2	60.00mil	Plated	+/-3.00mil
□	32	30.00mil	Plated	+/-3.00mil
☆	22	10.00mil	Plated	+/-2.00mil

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NOTES: UNLESS OTHERWISE SPECIFIED.

1. FABRICATE PER IPC-6012A CLASS 2.
2. FOR BOARD THICKNESS AND IMPEDANCE DETAILS REFER STACKUP DOCUMENT.
3. PRINTED WIRING BOARD SHALL COMPLY WITH REQUIREMENTS OF ANSI/J-STD-003.
4. SURFACE FINISH: IMMERSION SILVER
5. SOLDERMASK ON BOTH SIDES OF THE BOARD SHALL BE LPI. COLOR XXXXXX.
6. SILK SCREEN LEGEND TO BE APPLIED PER LAYER STACKUP USING WHITE NON-CONDUCTIVE EPOXY INK.
7. THIS PRINTED WIRING BOARD IS DESIGNED WITH A MINIMUM CONDUCTOR WIDTH AND SPACING OF 4 MIL & 4 MILS.
8. ALL VIAS ARE TENTED ON BOTH SIDES UNLESS SOLDERMASK OPENED IN GERBER.
9. ALL VIAS ON PAD SHOULD BE FILLED WITH NON CONDUCTIVE EPOXY AND SURFACE SHOULD BE FLAT. FLATNESS TOLERANCE FOR VIA ON PADS: +0.000/-0.001 INCHES ON BOTH SIDES. THE MANUFACTURER IS REQUESTED TO SIZE PER THEIR SOLDERMASK TOLERANCE.
10. SOLDER MASK OPENING IS KEPT SAME SIZE AS PAD (1:1) FOR ALL COMPONENTS
11. VENDOR SHOULD FOLLOW ROHS COMPLIANT PROCESS AND Pb FREE FOR MANUFACTURING
12. MANUFACTURER'S IDENTIFICATION. DATECODE LETTER SHALL BE SILKSCREENED ON SOLDER SIDE OF THE BOARD.
13. TRACE WIDTH SHOULD BE ACCURATELY ETCHED. MAX TOLERANCE +/- 1 MIL
14. ALL DIMENSIONS ARE IN MILS UNLESS OTHERWISE SPECIFIED.
15. FLATNESS REQUIREMENTS:
 - A. BOW AND TWIST OF ASSEMBLY SUB-PANEL OR SINGULATED PWB SHALL NOT EXCEED 0.7% OF LONGEST SIDE
 - B. TEST IN ACCORDANCE WITH THE CURRENT REVISION OF IPC-TM-650 2.4.22
16. PCB MATERIAL REQUIRMENTS:
 - A. FLAMMABILITY RATING MUST MEET OR EXCEED UL94V-0 REQUIREMENTS. PCB MUST BEAR THE UL94V-0 REGISTERED MATERIAL ID NUMBER.
 - B. Tg XXXXXX C OR EQUIVALENT.
 - C. EQUIVALENT MATERIAL SHALL BE RoHS COMPLIANT. HALOGEN FREE AND APPROVED BY ABC-Corp.
17. LAYER TO LAYER REGISTRATION SHALL BE WITHIN +/-2 MIL
18. FOR ETCHING TOLERANCE ON REF-DES XXXXX. REFER TO "XXXXX ETCHING

		UNLESS OTHERWISE SPECIFIED:		NAME	DATE	Sheild Board for Arduino UNO					
		DIMENSIONS ARE IN INCHES	DRAWN		07/03/2025						
		TOLERANCES:	CHECKED								
		FRACTIONAL±	ENG APPR.								
		ANGULAR: MACH± BEND ±	MFG APPR.								
		TWO PLACE DECIMAL ±	Q.A.			COMMENTS:					
		THREE PLACE DECIMAL ±									
		INTERPRET GEOMETRIC TOLERANCING PER:									
		MATERIAL				SIZE	DWG. NO.				
		FINISH									
NEXT ASSY	USED ON					SCALE: 1:1	WEIGHT:	SHEET 1 OF 3			
APPLICATION		DO NOT SCALE DRAWING									

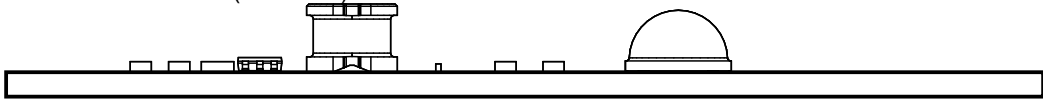
A

B

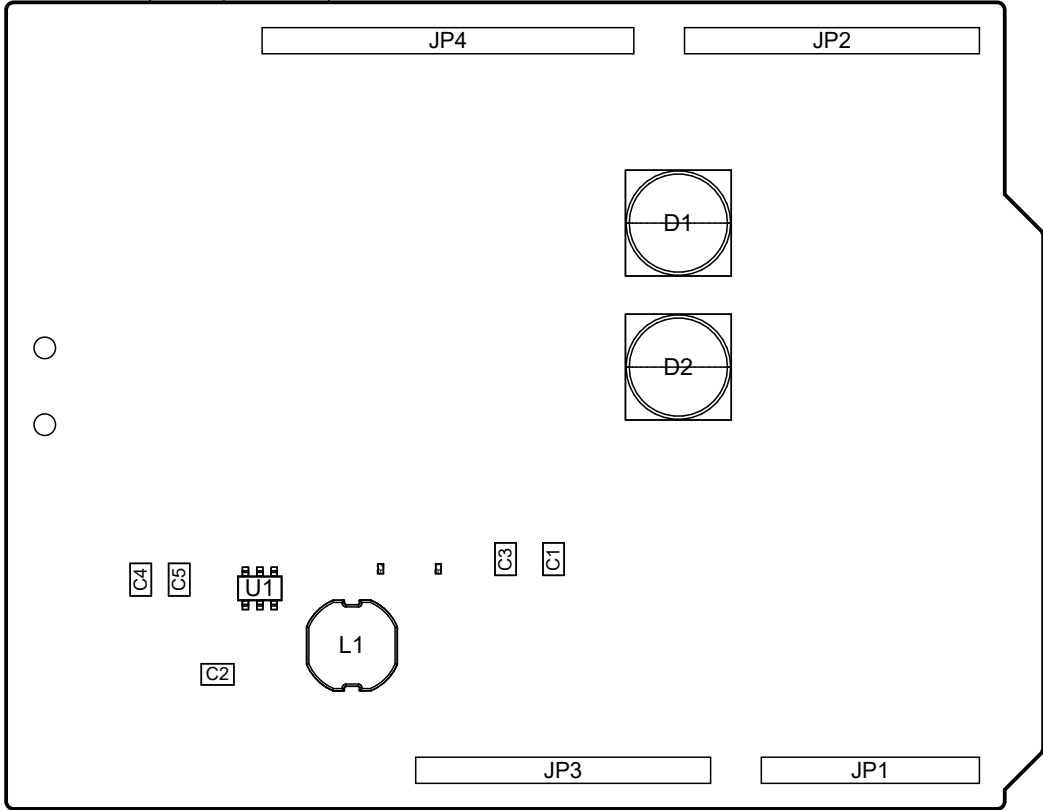
C

D

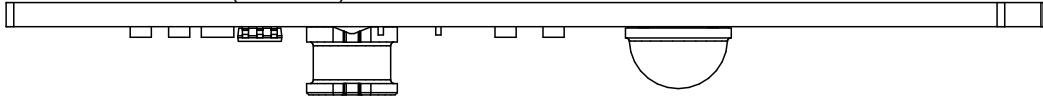
View from Front side (Scale 2:1)



View from Top side (Scale 2:1)



View from Back side (Scale 2:1)



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		UNLESS OTHERWISE SPECIFIED:		NAME	DATE	TITLE Sheild Board for Arduino UNO			
		DIMENSIONS ARE IN INCHES	DRAWN		07/03/2025				
		TOLERANCES:	CHECKED						
		FRACTIONAL±	ENG APPR.						
		ANGULAR: MACH± BEND ±	MFG APPR.						
		TWO PLACE DECIMAL ±	Q.A.			COMMENTS:			
		THREE PLACE DECIMAL ±							
		INTERPRET GEOMETRIC TOLERANCING PER:							
		MATERIAL				SIZE	DWG. NO.		
		FINISH							
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APPLICATION									

A

B

C

D

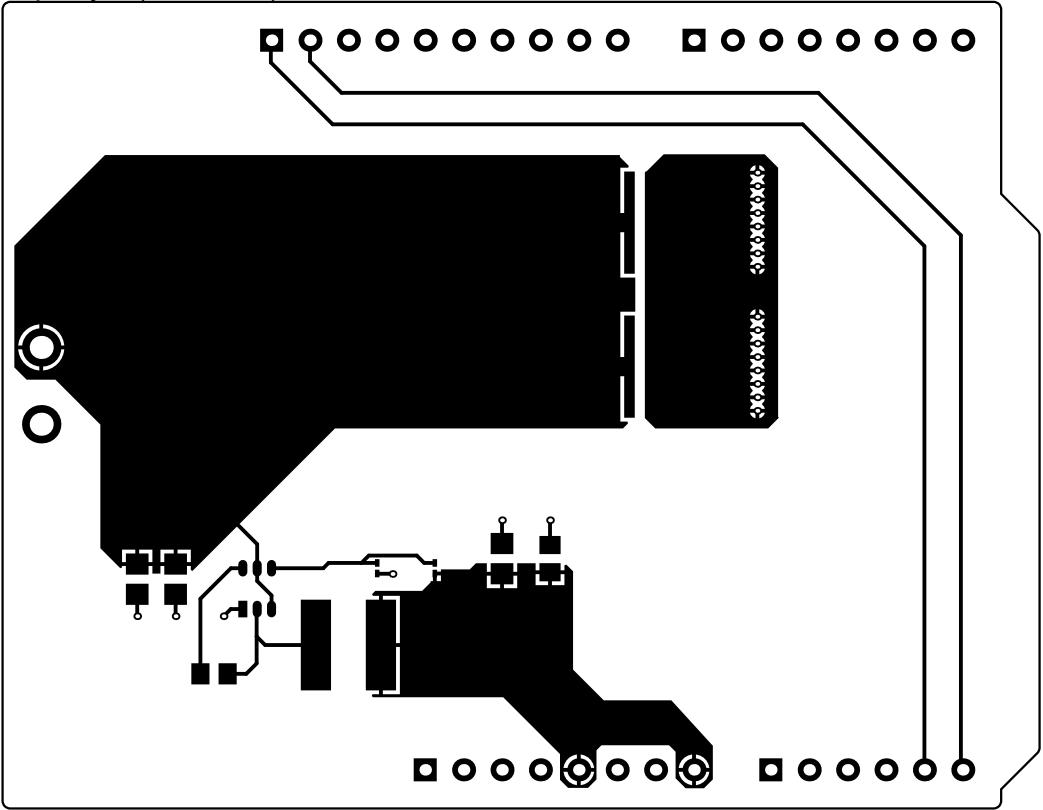
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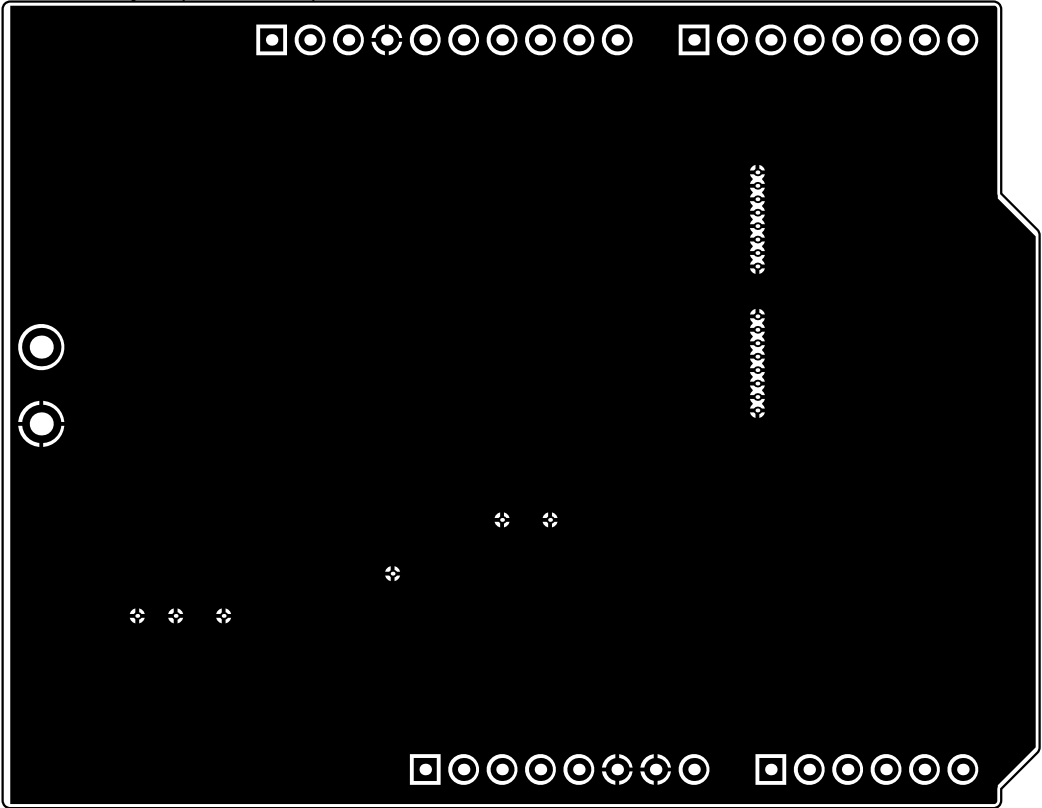
C

D

Top Layer (Scale: 2:1)



Bottom Layer (Scale: 2:1)



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		UNLESS OTHERWISE SPECIFIED:		NAME	DATE				
		DIMENSIONS ARE IN INCHES	DRAWN		07/03/2025				
		TOLERANCES:	CHECKED			TITLE Sheild Board for Arduino UNO			
		FRACTIONAL±	ENG APPR.						
		ANGULAR: MACH± BEND ±	MFG APPR.						
		TWO PLACE DECIMAL ±	Q.A.						
		THREE PLACE DECIMAL ±	COMMENTS:						
		INTERPRET GEOMETRIC							
		TOLERANCING PER:							
		MATERIAL				SIZE	DWG. NO.		
		FINISH							
NEXT ASSY	USED ON								
APPLICATION		DO NOT SCALE DRAWING				SCALE:	1:1	WEIGHT:	SHEET 3 OF 3

A

B

C

D

A

B

C

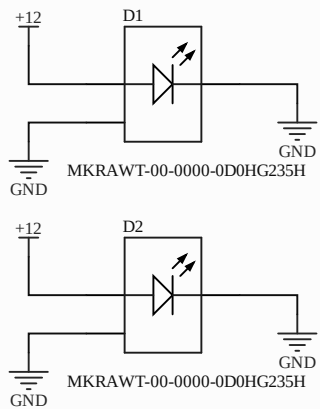
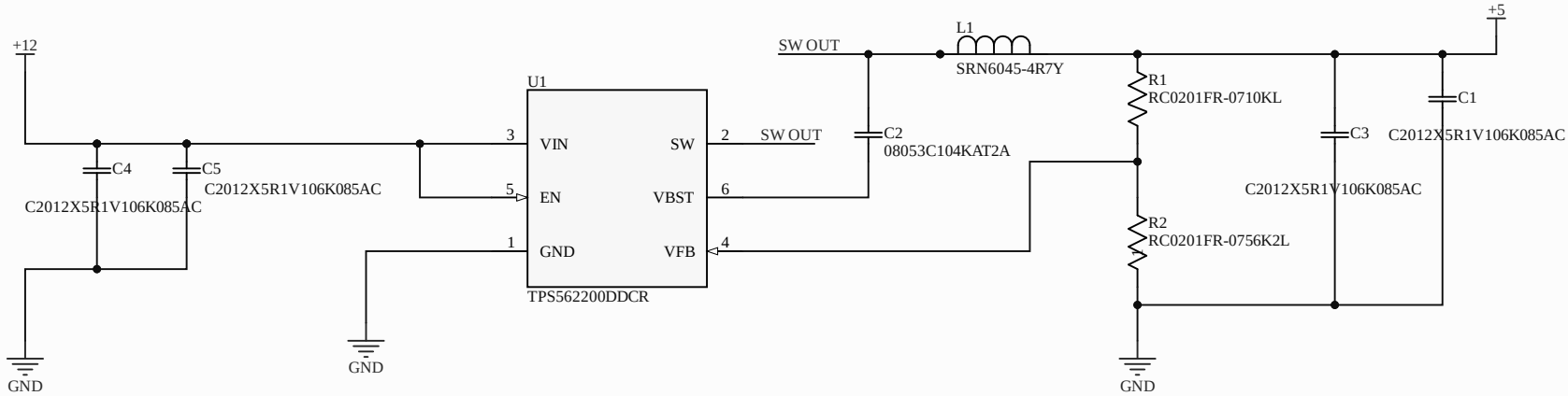
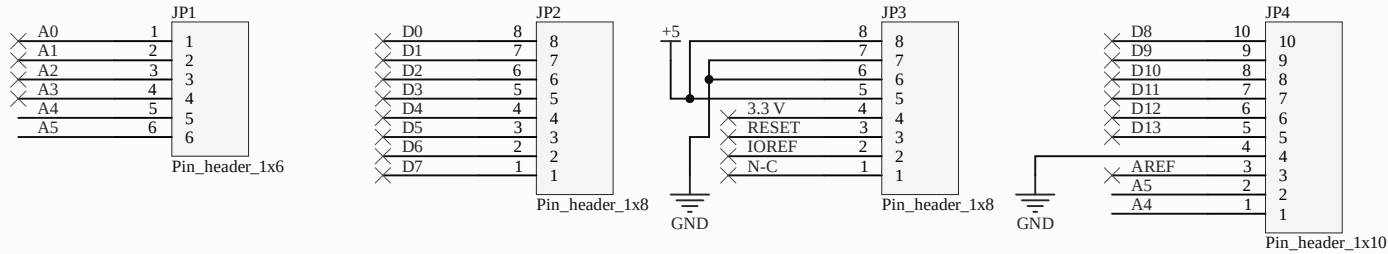
D

A

B

C

D

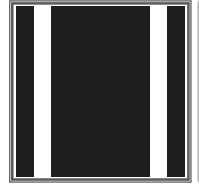


Title		
Power Regulator + LED Shield PCB for Arduino Uno		
Size	Number	Revision
A4	Drawing #0001	A
Date:	3/07/2025	Sheet of 1
File:	C:\Users\...\Schematic.SchDoc	Drawn By: Kavindu Makaranda

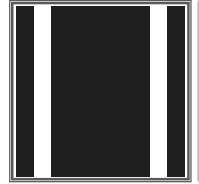
JP4



D1



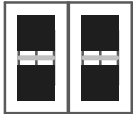
D2



PWR

GND

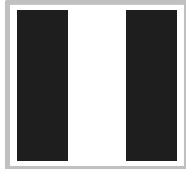
C4 C5



U1



L1



R2



R1



C3



C1



C2

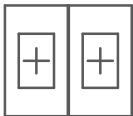
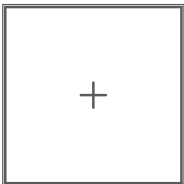
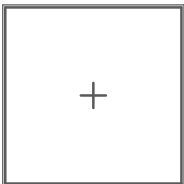


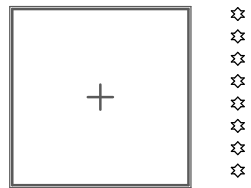
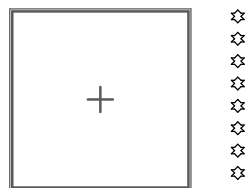
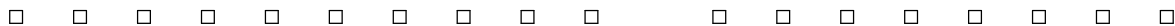
JP3



JP1

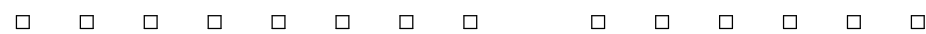
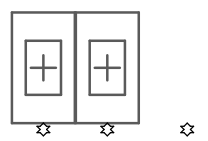






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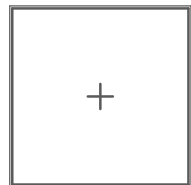
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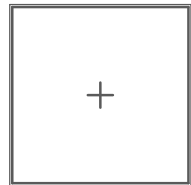
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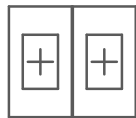
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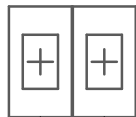
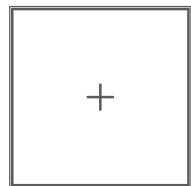
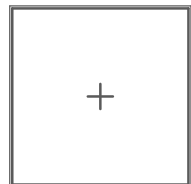
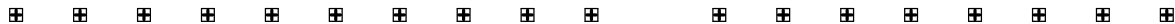


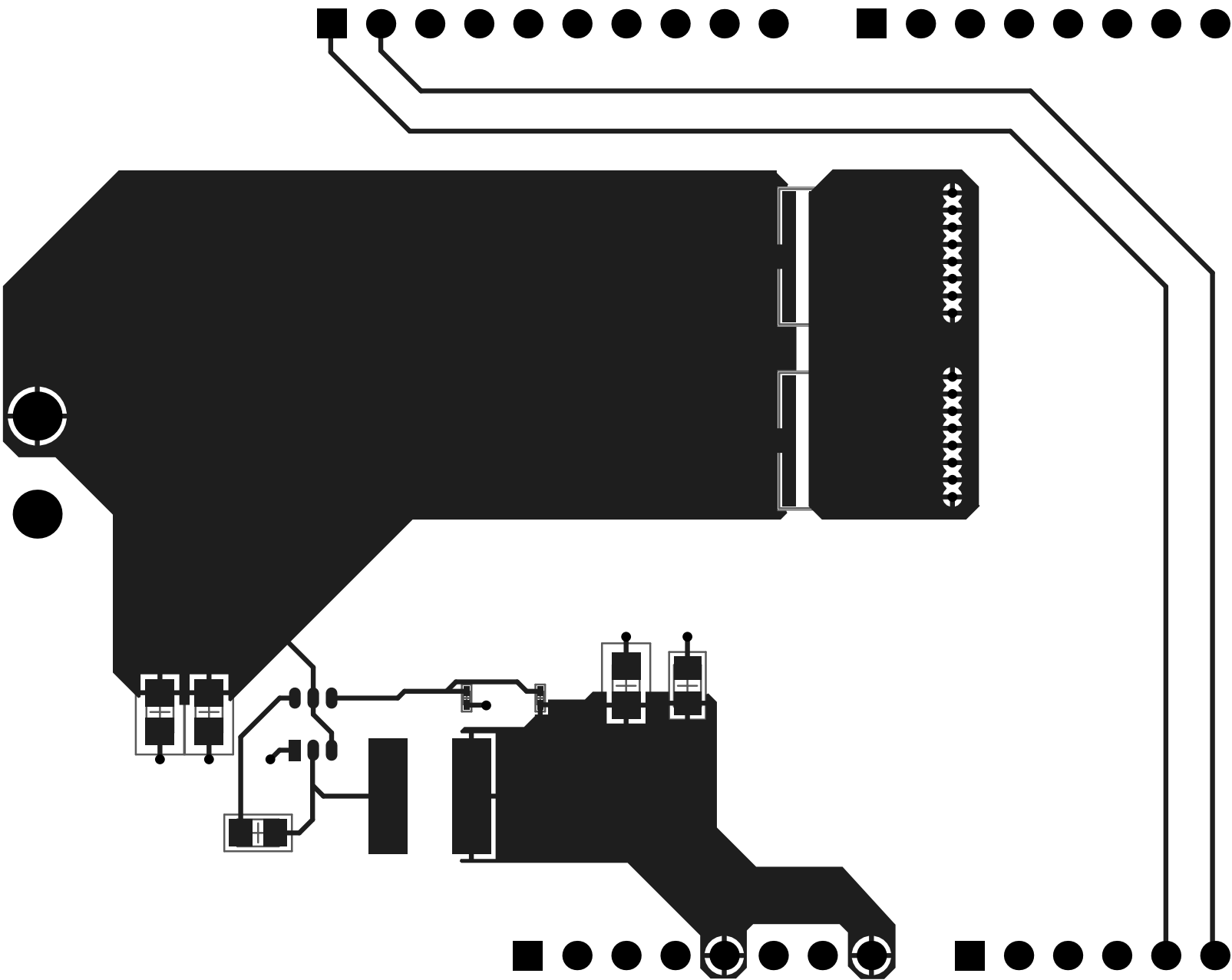
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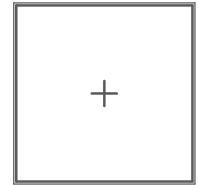




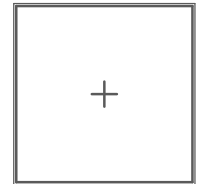
JP4

JP2

D1



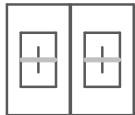
D2



PWR

GND

C4 C5



U1



R2



R1



L1



C3



C1

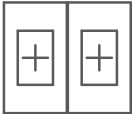
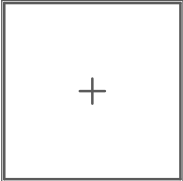


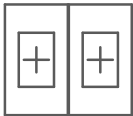
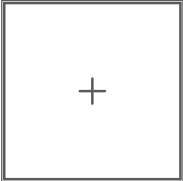
C2

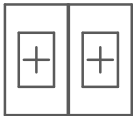
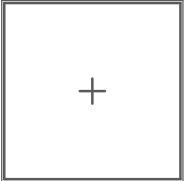


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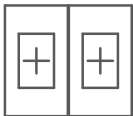
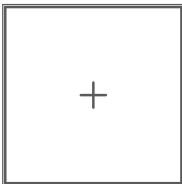
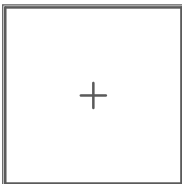
JP1

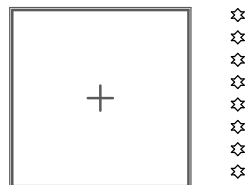
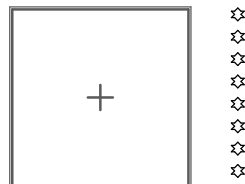
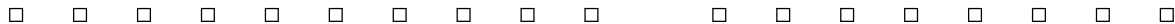






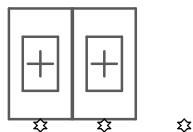






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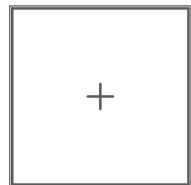
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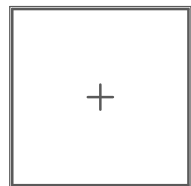
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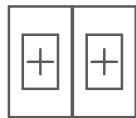
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| Line # | Name | Description | Designator | Revision ID | Revision State | Revision Status | Quantity | Manufacturer 1 | Manufacturer Part Number 1 | Manufacturer Lifecycle 1 | Supplier 1 | Supplier Part Number 1 | Supplier Unit Price 1 | Supplier Subtotal 1 |
|--------|--------------------------|---|----------------|-------------|----------------|-----------------|----------|-------------------|----------------------------|--------------------------------|--------------------|------------------------|-----------------------|---------------------|
| | C2012X5R1V106K085AC | C2012X5R1V106K085AC | C1, C3, C4, C5 | | Not managed | | 4 | TDK | C2012X5R1V106K085AC | Not Recommended for New Design | Digikey | 445-14417-1-ND | 0.42 | 1.68 |
| | 08053C104KAT2A | 08053C104KAT2A | C2 | | Not managed | | 1 | Kyocera AVX | 08053C104KAT2A | Volume Production | Avnet | 96M1381 | 0.114 | 0.114 |
| | MKRAWT-00-0000-0D0HG235H | LED, White, 12V, 2-Pin SMD, RoHS, Tape and Reel | D1, D2 | | Unknown server | | 2 | Cree | MKRAWT-00-0000-0D0HG235H | Volume Production | | | | |
| | Pin_header_1x6 | Pin_header_1x6 | JP1 | | Not managed | | 1 | | | | | | | |
| | Pin_header_1x8 | Pin_header_1x8 | JP2, JP3 | | Not managed | | 2 | | | | | | | |
| | Pin_header_1x10 | Pin_header_1x10 | JP4 | | Not managed | | 1 | | | | | | | |
| | SRN6045-4R7Y | No Description Available | L1 | | Not managed | | 1 | | | | | | | |
| | RC0201FR-0710KL | RC0201FR-0710KL | R1 | | Not managed | | 1 | Yageo | RC0201FR-0710KL | Volume Production | Future Electronics | 4031233 | | |
| | RC0201FR-0756K2L | No Description Available | R2 | | Not managed | | 1 | Yageo | RC0201FR-0756K2L | Volume Production | Future Electronics | 5109860 | 0.0013 | 13 |
| | TPS562200DDCR | No Description Available | U1 | | Not managed | | 1 | Texas Instruments | TPS562200DDCR | Volume Production | Mouser | 595-TPS562200DDCR | 0.66 | 0.66 |